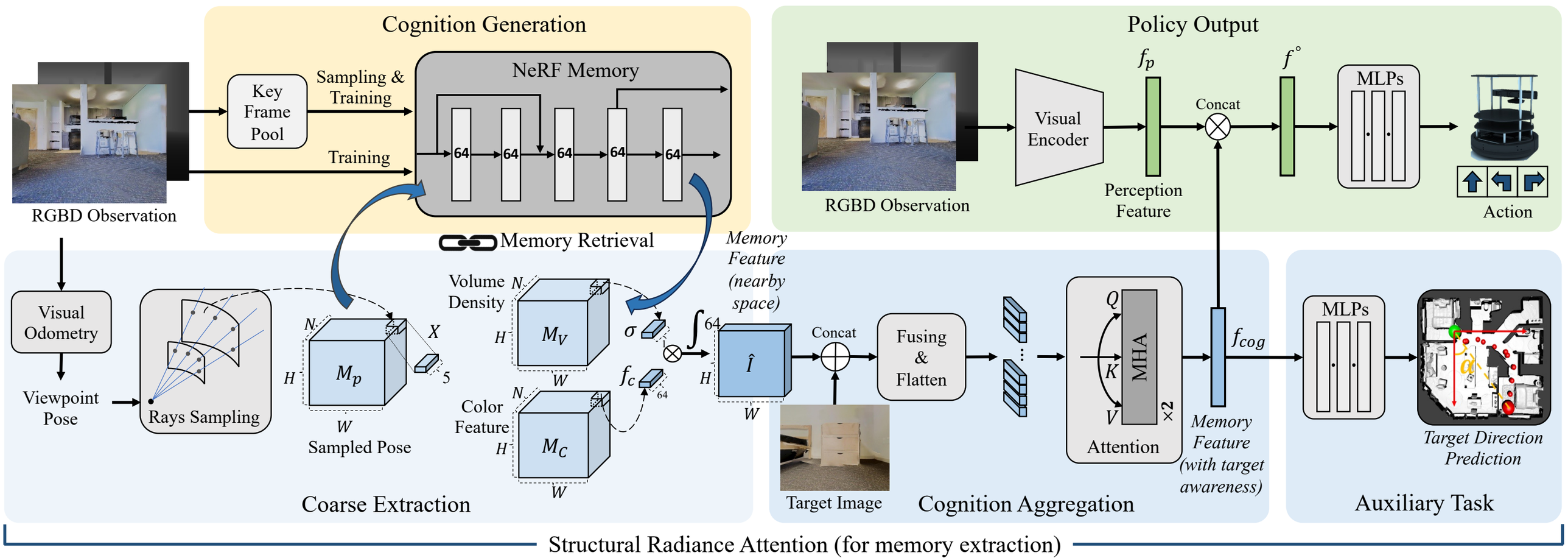
Cognition Generation
Policy Output
RGBD Observation
Key Frame Pool
Sampling & Training
Training
NeRF Memory
64
64
64
64
64
Memory Retrieval
Visual Encoder
f_p
Concat
f°
MLPs
Perception Feature
Action
Visual Odometry
Viewpoint Pose
Rays Sampling
N
H
W
M_p
X
5
Sampled Pose
Volume Density
M_V
σ
Color Feature
M_C
f_c
64
$\int$
$\hat{I}$
Memory Feature (nearby space)
Concat
Target Image
Fusing & Flatten
Q
K
V
MHA
×2
Attention
f_{cog}
Memory Feature (with target awareness)
MLPs
Target Direction Prediction
Coarse Extraction
Cognition Aggregation
Auxiliary Task
Structural Radiance Attention (for memory extraction)